R-UNIVERSALITY OF THE ABSOLUTELY SIMPLE ALGEBRAS OF TYPE E6: STRONG INNERNESS OR SPLITTING OVER AN IMAGINARY QUADRATIC EXTENSION

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Allowed standard prerequisites (editorial).

- The classification of forms of type E6 by their Tits index, and the notion of being strongly inner. Used inside the counted proof at source lines 796-800 and 818-820, attributed by the author to section 6.4.5 of the cited paper Tits-RepLinIrred. Supplies the list of strongly inner forms that the criteria are stated in terms of.
- The fact that the centre of a simply connected almost simple group of type E6 is cyclic of prime order. Used at the start of the counted proof (source lines 793-795). Reduces the criterion to a single Tits algebra, which is what makes the two alternatives exhaustive.

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Source reference. Original lemma 2.3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 471–471. Navigation pointer only: the author's lemma on the minimal inner extension; it has its own card in this lane and its content is declared as a named standard prerequisite.

Proposition 0.1. *Let $\mathfrak{g}$ be an absolutely simple Lie algebra over $\mathbb{Q}$ of type E_6. Then $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if either*

(1) *$\mathfrak{g}$ is strongly inner (that is, of type ${}^1E_{6,2}^{28}$ or ${}^1E_{6,6}^{0}$), or*
(2) *$\mathfrak{g}$ is an outer form that splits over an imaginary quadratic extension of $\mathbb{Q}$.*

Proposition 0.2 ([11, Corollary 3.5, § 4.2, and Lemma 7.4])**.** *Let L be the center of the algebra $D_{\mathfrak{g},F}(\lambda) := \operatorname{End}_{\mathfrak{g}}({}^F\rho_\lambda)$ (which, by Schur's lemma, is a division algebra). Then:*

(1) *$\operatorname{Gal}(\overline{F}/L) = \{\sigma \in \operatorname{Gal}(\overline{F}/F) \mid \sigma(\lambda) = \lambda\}$ (for the $*$-action of $\operatorname{Gal}(\overline{F}/F)$), and*
(2) *$\beta_{\mathfrak{g},L}(\lambda) = [D_{\mathfrak{g},F}(\lambda)]$, after identifying $H^2(L;\boldsymbol{\mu})$ with the Brauer group of L.*

Proof. Assume, first, that $\mathfrak{g}$ is inner. Since the center of any simply connected, almost simple group of type E_6 is cyclic of prime order (namely, it is of order 3), we see that $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if either $\mathfrak{g}$ is strongly inner (over $\mathbb{Q}$) or $\mathfrak{g} \otimes_{\mathbb{Q}} \mathbb{R}$ is not strongly inner. From [11, § 6.4.5], we see that Lie algebras of type ${}^1E_{6,2}^{28}$ or ${}^1E_{6,6}^{0}$ are always strongly inner, and those of type ${}^1E_{6,2}^{16}$ are never strongly inner. In addition, a Lie algebra of type ${}^1E_{6,0}^{78}$ over $\mathbb{Q}$ cannot be strongly inner [12, Propositions 4 and 5]. Since $\mathfrak{g} \otimes_{\mathbb{Q}} \mathbb{R}$ must be of type ${}^1E_{6,2}^{28}$ or ${}^1E_{6,6}^{0}$, it is strongly inner. Hence, $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if it is strongly inner.

Suppose, now, that $\mathfrak{g}$ is outer, and let L be the unique quadratic extension of $\mathbb{Q}$ over which $\mathfrak{g}$ is inner. From Lemma 2.3, we know that if $\mathfrak{g}$ is $\mathbb{R}$-universal, then it must remain outer over $\mathbb{R}$, so L is an imaginary extension of $\mathbb{Q}$. Let λ be a weight that represents a nontrivial element of Z^*. Then λ is not fixed by the $*$-action, so $D_{\mathfrak{g},\mathbb{R}}(\lambda) = D_{\mathfrak{g},\mathbb{C}}(\lambda) = \mathbb{C}$ and $D_{\mathfrak{g},\mathbb{Q}}(\lambda) = D_{\mathfrak{g},L}(\lambda)$ (see Proposition 0.2). So $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if $\mathfrak{g} \otimes_{\mathbb{Q}} L$ is strongly inner.

Since L is an imaginary extension, we know that $\mathfrak{g}$ splits at the infinite place of L. Then, by inspection of the possible Tits indices of type ${}^1E_6$ over a nonarchimedean local field [10, p. 58], we see that the central vertex of the Tits index is circled at every place, so it must be circled over L [2, Satz 4.3.3]. Therefore, $\mathfrak{g}$ must be either split or of type ${}^1E_{6,2}^{16}$ over L. From [11, § 6.4.5], we see that the form of type ${}^1E_{6,2}^{16}$ is not strongly inner. Therefore, $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if $\mathfrak{g} \otimes_{\mathbb{Q}} L$ is split. □

References

[1] Demazure M., Grothendieck A., Schémas en groups. III. Structure des schémas en groupes reductifs, Séminaire de géométrie algébrique du Bois Marie, 1962/1964, available at `http://library.msri.org/books/sga/sga/pdf/sga3-3.pdf`.

[2] Harder G., Bericht über neuere Resultate der Galoiskohomologie halbeinfacher Gruppen, *Jber. Deutsch. Math.-Verein.* **70** (1968), 182–216, available at `http://www.digizeitschriften.de/en/dms/img/?PPN=PPN37721857X_0070&DMDID=dmdlog17`.

[3] Kneser M., Lectures on Galois cohomology of classical groups, *Tata Institute of Fundamental Research Lectures on Mathematics*, Vol. 47, Tata Institute of Fundamental Research, Bombay, 1969, available at `http://www.math.tifr.res.in/~publ/ln/tifr47.pdf`.

[4] Knus M.A., Merkurjev A., Rost M., Tignol J.P., The book of involutions, *American Mathematical Society Colloquium Publications*, Vol. 44, Amer. Math. Soc., Providence, RI, 1998.

[5] Lam T.Y., Introduction to quadratic forms over fields, *Graduate Studies in Mathematics*, Vol. 67, Amer. Math. Soc., Providence, RI, 2005.

[6] Morris D., Real representations of semisimple Lie algebras have $\mathbb{Q}$-forms, in Algebraic Groups and Arithmetic, Tata Institute of Fundamental Research, Mumbai, 2004, 469–490.

[7] Platonov V., Rapinchuk A., Algebraic groups and number theory, *Pure and Applied Mathematics*, Vol. 139, Academic Press, Inc., Boston, MA, 1994.

[8] Prasad G., Rapinchuk A., On the existence of isotropic forms of semi-simple algebraic groups over number fields with prescribed local behavior, *Adv. Math.* **207** (2006), 646–660.
[9] Raghunathan M.S., Arithmetic lattices in semisimple groups, *Proc. Indian Acad. Sci. Math. Sci.* **91** (1982), 133–138.
[10] Tits J., Classification of algebraic semisimple groups, in Algebraic Groups and Discontinuous Subgroups (Proc. Sympos. Pure Math., Boulder, Colo., 1965), Amer. Math. Soc., Providence, R.I., 1966, 33–62.
[11] Tits J., Représentations linéaires irréductibles d'un groupe réductif sur un corps quelconque, *J. Reine Angew. Math.* **247** (1971), 196–220.
[12] Tits J., Strongly inner anisotropic forms of simple algebraic groups, *J. Algebra* **131** (1990), 648–677.